

Pouring the Future

A joint research partnership between the Human-Technology Interaction Lab (University of Alabama) and the Computing for Social Good Lab (University of Florida).

A large, bold, red graphic of the text "\$1.1B" is positioned on the left side of the page. The dollar sign is stylized with a vertical line through it. The numbers are also bold and red.

Coca-Cola's announced investment in artificial intelligence.

Coca-Cola is putting \$1.1B into AI. The open question is human.

The beverage, the machine and the moment of choice are all being re-invented. The human science is still missing — how people feel, learn, and are treated fairly by the systems that serve them.

WHY NOW

Four frontiers, already in motion.

01

Personalization

AI-driven drinks — Share a Coke on Freestyle.

02

Connected dispensing

Next-gen Freestyle Equinox, Mini, mixology counters.

03

Sustainability

World Without Waste and reverse vending.

04

Inclusive access

Serving every body, ability and language.

- Each frontier raises a human question that in-house engineering is not built to answer.

THE PAIRING

Two labs, one deliberate contrast.

HTIL · UNIVERSITY OF ALABAMA

Dr. Chris S. Crawford

Focus: the body as an interface.

Brain–computer interfaces and human–robot interaction

Reads human state in real time: attention, arousal, affect

NeuroBlock — low-code neurofeedback for beginners

Low-cost physiological sensing hardware (~\$4 BOM)

NSF CAREER award

CFSG · UNIVERSITY OF FLORIDA

Dr. Juan E. Gilbert

Focus: technology that serves everyone, fairly.

National Medal of Technology and Innovation, 2023 · ACM
Fellow

Prime III — open-source universal-design voting, used in
real elections

Applications Quest — patented AI for fairer, holistic
decisions

Equity and bias in AI, accessibility

Culturally relevant computing and learning technology

Four capabilities Coca-Cola cannot buy off the shelf.

HUMAN-STATE SENSING

Physiological and affective computing, from the electrodes up.

UNIVERSAL DESIGN

Accessibility as the design premise, not a retrofit.

EQUITABLE AI

Auditing and correcting models for bias across people and places.

LEARNING SYSTEMS

Intelligent tutoring and low-code tools that teach at scale.

-
- Together: the human layer of Coca-Cola's digital future.

THE PORTFOLIO

Six projects. One partnership.

A Every Customer, Every Ability	Inclusive access
B The Responsive Pour	Personalization
C Fair by Design	Responsible AI and trust
D Every Bottle Back	Sustainability
E The Flavor Academy	Digital capability and workforce
F The Living Lab	Brand experience

Each is a research question, not a spec — designed to produce peer-reviewed results and protectable IP.

PROJECT A · INCLUSIVE ACCESS

Every customer, every ability.

WHAT WE BUILD

Prime III's universal-design method applied to next-gen dispensers — tactile, audio, haptic and multilingual interaction, usable by low-vision, blind, motor-impaired and first-time users without a phone.

WHY IT MATTERS TO COCA-COLA

Inclusive-by-design equipment widens the addressable market, answers tightening law, and turns "accessible" from a compliance cost into a brand strength.

The responsive pour.

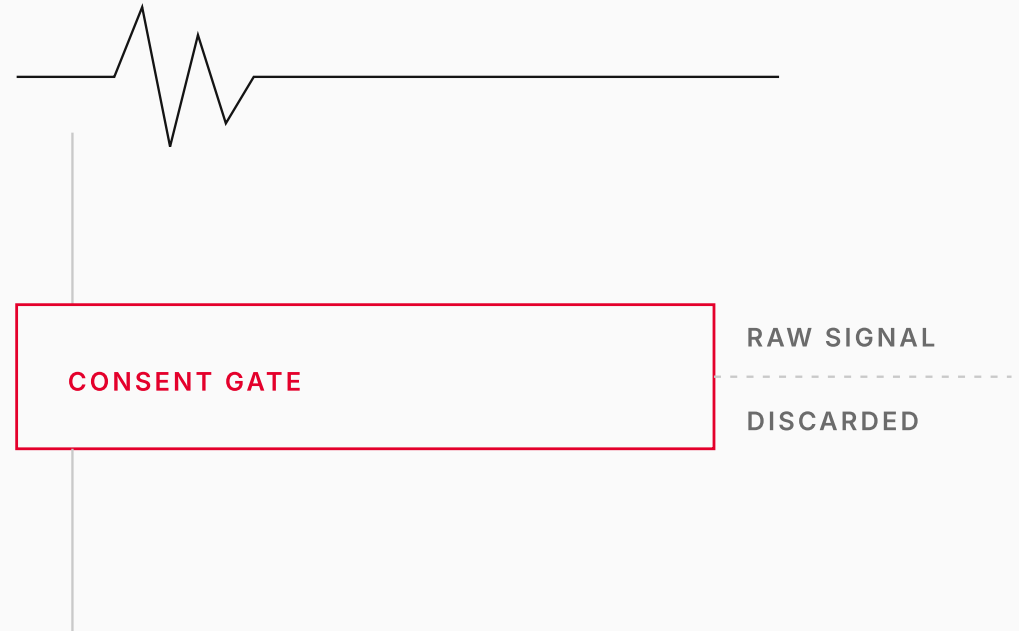
WHAT WE BUILD

Dispensing that senses engagement and arousal — never identity — and adapts the Share-a-Coke moment in real time, inside a consent-first, privacy-preserving architecture.

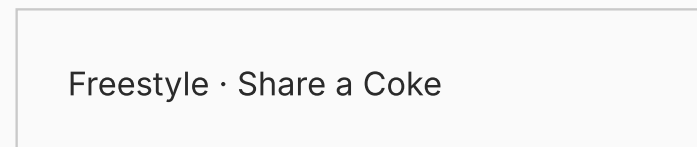
WHY IT MATTERS

Personalization that respects the person: higher experiential resonance, without the backlash of covert sensing.

ENGAGEMENT · AROUSAL

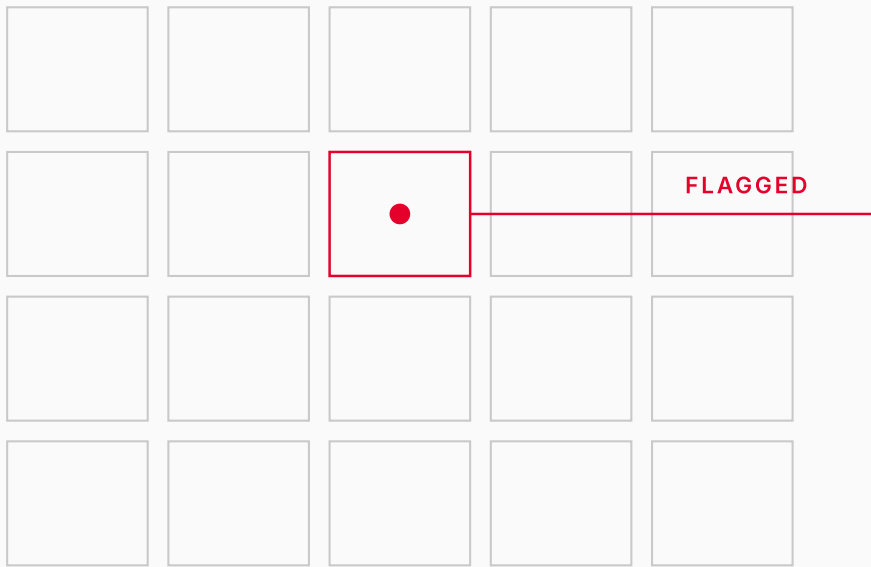


ADAPTED MOMENT



Fair by design.

MODEL × SEGMENT AUDIT



DEMOGRAPHICS · REGIONS · STORE TYPES

WHAT WE BUILD

Applications Quest adapted into an independent bias-audit framework for personalization, pricing and marketing models — measured across demographics, regions and store types, with reproducible metrics.

WHY IT MATTERS

Credible, publishable assurance that a \$1.1B AI investment treats consumers equitably — and a method that catches bias before it becomes a headline.

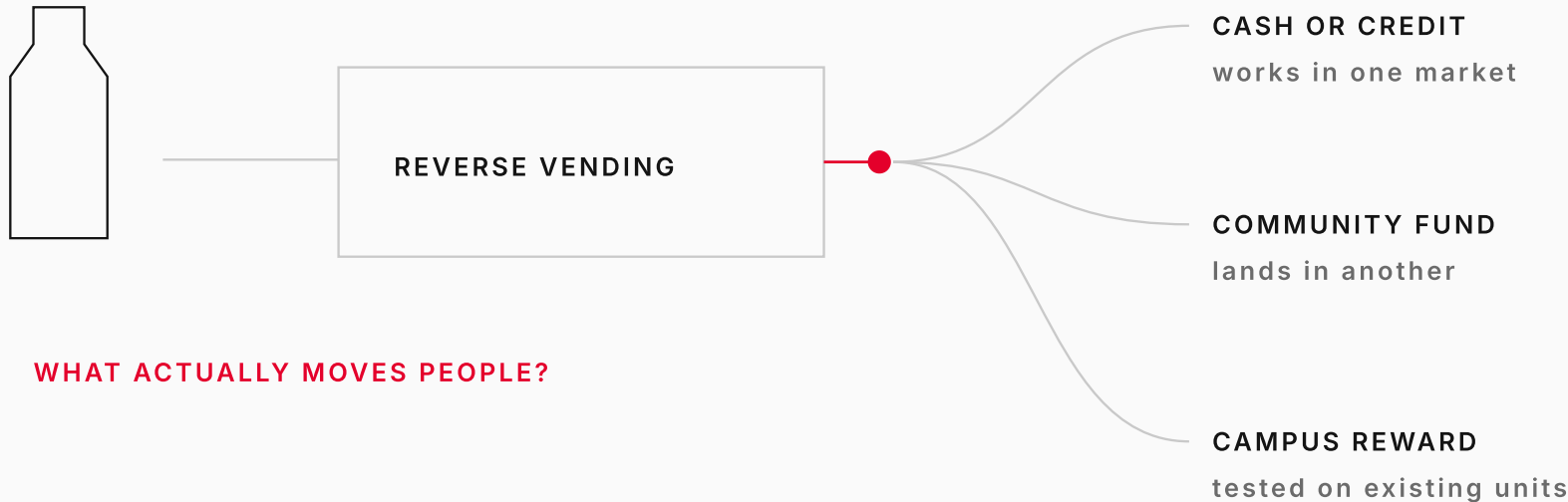
Every bottle back.

WHAT WE BUILD

Behavioral science and culturally relevant design layered onto reverse vending — tested on the existing footprint, including units already running on the University of Alabama campus.

WHY IT MATTERS

World Without Waste needs behavior change, not just machines. We test what moves people, community by community, and hand back a playbook.



PROJECT E · LEARNING AND WORKFORCE

The flavor academy.

WHAT WE BUILD

Two learning platforms from one engine — an intelligent-tutoring system for bottler and franchise teams, and a consumer-facing "science of taste" experience — both authored with the low-code model proven by NeuroBlock.

WHY IT MATTERS TO COCA-COLA

Faster, cheaper workforce readiness across the bottling system, plus a branded learning experience that deepens consumer connection and feeds a STEM pipeline.

PROJECT F · BRAND EXPERIENCE

The living lab.

WHAT WE BUILD

An instrumented experience at Coca-Cola-sponsored venues — real-time engagement measured with physiological sensing, plus embodied hosts for accessible hospitality and wayfinding.

WHY IT MATTERS

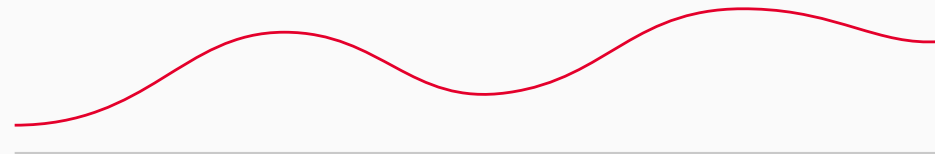
Sponsorship becomes measured experience science, with findings that feed straight back into products.

CROWD



SENSING · EMBODIED HOSTS

MEASURED ENGAGEMENT



THE OFFER

Not a vendor. A research partner.

PUBLISHABLE, PATENTABLE IP

Peer-reviewed results and protectable technology, co-owned under a negotiated agreement.

DE-RISKED INNOVATION

Small, evidence-backed pilots before any large commitment.

CREDIBILITY

Accessibility, equity and sustainability claims backed by independent university research.

TALENT

A direct pipeline to trained graduate researchers in HCC, AI and BCI.

THE MODEL

How we would work together.

90-DAY SCOPING SPRINT



Pick the first two projects

FUNDED PILOT YEAR



Evidence, prototypes, papers

ANNUAL REVIEW



Choose what scales

STRUCTURE

Two PIs, one program manager, 4–6 graduate researchers across both campuses.

FUNDING

Sponsored research matched by federal and foundation grants, lowering net cost per project.

DELIVERABLES

Quarterly findings, published papers, working prototypes, annual leadership report.

The drink is already engineered to a science. The moment around it is **not — yet.**